

Exploratory Diagnostics on Supply Chain Vulnerabilities and E-Commerce Returns

Step 1: Project Introduction

The main aim of this project is to analyze supply chain problems in an e-commerce business. The project focuses on identifying product returns, delivery delays, warehouse inefficiencies, and seasonal logistics issues.

The analysis helps businesses understand why customers return products and which factors affect delivery performance.

Step 2: Dataset Collection

The Olist Brazilian E-Commerce Public Dataset was used for this project.

The following datasets were collected:

- Orders Dataset (orders.csv)
- Order Items Dataset (order_items.csv)
- Products Dataset (products.csv)
- Customers Dataset
- Sellers Dataset
- Reviews Dataset
- Payments Dataset
- Geolocation Dataset

These datasets contain information about orders, products, sellers, shipping costs, and customer locations.

Step 3: Loading the Dataset

The first step in coding was loading the required datasets into Python using the Pandas library.

Example:

```
import pandas as pd

orders = pd.read_csv("orders.csv")
items =
pd.read_csv("order_items.csv")
products =
```

Purpose:

- Read the data files.
 - Convert them into DataFrames.
 - Prepare data for analysis.
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Step 4: Data Merging

After loading the datasets, they were merged together.

```
df = orders.merge(items,  
on='order_id') df = df.merge(products,  
on='product_id')
```

Purpose:

- Combine order information with product information.
 - Create one complete dataset for analysis.
 - Enable easier exploration of relationships between orders and products.
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Step 5: Data Understanding

The merged dataset contains:

- Order details
- Product category
- Seller information
- Freight cost
- Delivery information

The dataset contains more than 100,000 transactions.

Purpose:

- Understand data structure.
- Identify important columns for analysis.

Step 6: Data Handling

The dataset was checked for missing values and duplicate records to improve data quality and ensure accurate analysis.

Purpose:

- Identify missing values in the dataset.
- Detect and remove duplicate records.
- Improve data consistency and reliability.
- Prepare clean data for analysis and modelling.

Activities Performed:

- Checked null values using `isnull().sum()`.

- Identified duplicate records using duplicated().sum().
 - Removed unnecessary duplicates.
 - Verified data quality before further analysis.
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Step 6: Exploratory Data Analysis (EDA)

EDA was performed to understand patterns and trends in the dataset.

Analysis 1: Highest Return Categories

Objective: Identify product categories with the highest return rates.

Method:

- Count returned products by category.
- Create bar charts and count plots.

Outcome:

- Find categories responsible for most returns.
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Analysis 2: Delivery Delay Analysis

Objective: Identify warehouses or sellers with the longest delays.

Formula:

Delivery Time = Delivery Date - Purchase Date

Purpose:

- Measure shipping efficiency.
 - Detect delay-prone regions.
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Analysis 3: Monthly Return Trend

Objective: Find months with the highest return volume.

Method:

- Extract month from purchase date.
- Create monthly trend charts.

Purpose:

- Identify seasonal return patterns.
- Understand peak return periods.

Analysis 4: Freight Cost Impact

Objective: Determine whether shipping cost influences returns.

Method:

- Scatter plot analysis.
- Compare freight value with return frequency.

Purpose:

- Understand the relationship between shipping cost and customer satisfaction.
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Step 10: Data Visualization

Different charts were created to present findings.

Basic Visualizations

- Histogram
- Bar Chart
- Pie Chart

Intermediate Visualizations

- Box Plot
- Violin Plot
- Heatmap

Advanced Visualizations

- Delivery Delay Dashboard
- Geographic Delivery Analysis

Purpose:

- Present results visually.
 - Make business insights easier to understand.
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Step 11: Key Insights Generated

The project aims to identify:

1. Highest-return product category.

2. Most delayed region.
3. Most efficient seller.
4. Peak return season.

These insights help improve supply chain performance.

Step 12: Dashboard Creation

A Return Analytics Dashboard was developed.

KPIs Included:

- Total Orders
- Return Rate
- Delivery Delay Percentage

Purpose:

- Provide quick business insights.
 - Support decision-making.
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Step 13: Predictive Modelling

A Delivery Delay Prediction model was developed.

Algorithms Used:

- Logistic Regression
- Random Forest

Output:

- Delayed = Yes or No

Purpose:

- Predict future delivery delays.
 - Reduce customer complaints.
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Step 14: Supply Chain Risk Scoring

Risk factors considered:

- Product Category
- Seller
- Region
- Freight Cost

Output Levels:

- Low Risk
- Medium Risk
- High

Risk. Purpose:

- Identify risky supply chain operations.
 - Improve logistics planning.
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Step 15: Conclusion

This project successfully analyzed e-commerce supply chain performance using real-world transaction data. Through data loading, cleaning, merging, visualization, statistical analysis, and machine learning techniques, important business insights were generated regarding product returns and delivery delays.

The findings can help organizations reduce logistics costs, improve customer satisfaction, optimize warehouse operations, and strengthen overall supply chain efficiency.

